HARD DISK

Hard disk drives are one of the most common choke points in your system. Find out what makes them walk or zoom

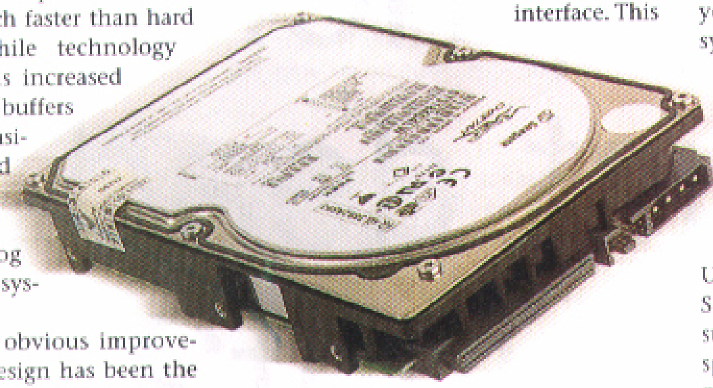
When it comes to storing large quantities of data and still keeping the cost per MB down to a few paise, there is no other storage device like the hard disk drive. Born out of a marriage of fields as diverse as electronics, mechanics, nanotechnology, aerodynamics and even chemical engineering, hard disk drives are true triumphs of modern engineering.

Components such as processors and RAM shuttle data much faster than hard disk drives, and while technology advancements such as increased spindle speeds, larger buffers and greater storage densities have improved hard disk performance, they are still the devices that can bog down the rest of your system.

One of the most obvious improvements in hard disk design has been the increase in spindle speed. The transition from 5,400 to 15,000 rpm has ensured that hard disks can meet the expectations of demanding applications such as image and video editing. For example, you may have noticed that your hard disk's LED indicator blinks wildly while you were playing *Soldier of Fortune 2*. This is because today's games with their bandwidth-sucking video cutscenes, huge virtual worlds and megabyte-sized character textures stress on the sequential and random read/write capabilities of your hard disk. Ideally, 7,200-rpm drives are best suited for home and office applications. If you have a slower hard disk, you will lose out on significant performance boosts, whether it is in terms of faster boot up times and application launches or smoother gaming.

Another very important component of your hard disk is the onboard buffer. This is the memory which holds the information that is being read or written on to the hard disk's platters. Today's hard disks come with anything from 2 MB to 8 MB of buffer memory. Since the buffer comes into play during the entire operation of your hard disk, a drive with a larger buffer will exhibit speed increases across the board.

One of the most neglected features is the interface of the hard disk. Almost all of today's IDE hard disks are of the ATA/100 and ATA/133 variety. So when you finally buy that new 7,200-rpm drive, the performance jump will be noticeable only if it is coupled with the right interface. This



A hard disk's rotations per minute (rpm) is the prime reason for speed. Drives with higher rpm, say 10,000 rpm, will supercharge your data-intensive applications

means that the data can be transferred at a theoretical rate of 100 MBps in the case of ATA/100 and 133 MBps in the case of ATA/133.

Apart from the fact that these rates are never really seen in real life (practically optimum rates are in the range of 32 to 36 MBps in the fastest 7,200-rpm drives), if you do not use the proper cables to connect these hard disks to your system, you will lose out on all the performance benefits associated with new hard disks. You will notice the speed improvements only if you use a compliant 80-wire, 40-pin cable. Of course, your motherboard needs to support the same ATA standard that your hard disk is using! Using these new hard disks on an older system such as a 440BX-based motherboard would completely throttle its performance. You will also need to add an additional IDE card that features this new interface if you want to see the associated performnace increases with newer hard disks.

On the higher end of the scale, SCSI hard disks offer further increases in data transfer speeds and they also break past IDE's restriction of having a maximum of two hard disks per channel—SCSI allows a maximum of 15 devices on a single chain. However, even though you have a SCSI hard disk in your system, the response of your applications may not be much higher than that of a new system with one of the new IDE hard disks as newer IDE hard disks have managed to surpass the data transfer rates of some of the older flavrous of SCSI such as Wide SCSI and Fast SCSI. Unless you have one of the newer SCSI hard disks based upon standards such as Ultra160 boasting of 10,000 rpm speeds, you would not really perceive much of a speed difference compared to some of today's ATA/133 7,200-rpm IDE drives. Also, SCSI hard disks are more expensive to implement compared to IDE and with the enhanced expansion capabilities of the SCSI standard, they are better suited to corporate applications for use in workstations and servers.

Tips & Tweaks

- Regular scanning and defragmentation of your hard disk keeps it healthy and enables it to operate at optimum levels.
- Enable the DMA setting on your hard disk within Windows if it isn't already enabled.
- Set the PIO/DMA setting in the BIOS to DMA for all your hard disks.
- Using two hard disks together with a RAID controller card configured in RAID-0 will deliver up to 40 per cent increase in the data transfer speed.